

EE 5393 - Circuits, Computation, and Biology

Homework #1 Report

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AI Disclosure:

At the beginning of this assignment, I used AI (Claude) to help me understand the provided *alea/vanilla* simulator codebase (<http://mriedel.ece.umn.edu/files/aleae.zip>). This allowed me to learn how the code was structured, how it functions, and how to correctly run and work with it on my machine.

After developing a clear understanding of the system, I again used Claude to assist with writing the C++ code. However, for each problem, I provided specific instructions and guidance in my prompts based on my own understanding of the assignment requirements. In this sense, the AI served as a coding assistant, while the design decisions and problem-solving approach were driven by my own comprehension.

Link to my GitHub Repository : https://github.com/mbinte/EE-5393_Spring_2026_HomeWorks.git

Problem 1: Analyzing Chemical Reaction Networks

1.1 Background Theory

Chemical reaction networks (CRNs) model systems where molecular species interact according to defined rules. Each reaction specifies reactants, products, and a rate constant. There are two complementary modeling paradigms:

Continuous (deterministic) model: Molecular concentrations evolve according to ordinary differential equations (ODEs). The rate of each reaction is proportional to the product of reactant concentrations raised to their stoichiometric coefficients, multiplied by the rate constant. This corresponds to the **mass-action kinetics** assumption.

Discrete (stochastic) model: Individual molecule counts evolve through probabilistic discrete events, following the **Gillespie algorithm** (Stochastic Simulation Algorithm, SSA). At each step:

1. Compute the propensity of each reaction: $a_i = k_i \times C(x_1, n_1) \times C(x_2, n_2) \times \dots$, where $C(x,n)$ is the binomial coefficient " x choose n "
2. The total propensity is $A = \sum a_i$
3. The time to the next reaction is an exponential random variable with mean $1/A$
4. The probability of reaction i firing is a_i / A

The key difference between the two models: in the continuous model, the rate is proportional to x^n (concentration raised to stoichiometric coefficient), while in the discrete model, propensity uses $C(x,n) = x!/(n!(x-n)!)$, accounting for the combinatorial ways molecules can interact.

1.2 The Reaction System

The reactions under study are:

Reaction	Rule	Rate Constant
R1	$2X_1 + X_2 \rightarrow 4X_3$	$k_1 = 1$
R2	$X_1 + 2X_3 \rightarrow 3X_2$	$k_2 = 2$
R3	$X_2 + X_3 \rightarrow 2X_1$	$k_3 = 3$

The discrete propensities are:

- $a_1 = k_1 \times C(x_1, 2) \times C(x_2, 1) = \frac{1}{2} x_1(x_1 - 1) x_2$
- $a_2 = k_2 \times C(x_1, 1) \times C(x_3, 2) = x_1 x_3(x_3 - 1)$
- $a_3 = k_3 \times C(x_2, 1) \times C(x_3, 1) = 3 x_2 x_3$

Problem 1(a): Outcome Probabilities

Setup: Starting from state $S = [X_1=110, X_2=26, X_3=55]$, the system evolves stochastically until one of the following outcomes is first reached:

- **C1:** $X_1 \geq 150$
- **C2:** $X_2 < 10$
- **C3:** $X_3 > 100$

Method: 100,000 independent stochastic simulation trials were performed using the Gillespie algorithm. Each trial began from the initial state and ran until one of the three threshold conditions was satisfied.

Results:

Outcome	Condition	Count	Probability
C1	$X_1 \geq 150$	0	$\approx 0.00\%$
C2	$X_2 < 10$	0	$\approx 0.00\%$
C3	$X_3 > 100$	100,000	$\approx 100.00\%$

Average events per trial: approx. 1,985

Analysis:

The system overwhelmingly reaches the C3 outcome ($X_3 > 100$). To understand why, consider the initial propensities from $S = [110, 26, 55]$:

- $a_1 = \frac{1}{2} \times 110 \times 109 \times 26 = 155,870 \rightarrow P(R1) \approx 32.0\%$
- $a_2 = 110 \times 55 \times 54 \times (2/2) = 326,700 \rightarrow P(R2) \approx 67.1\%$
- $a_3 = 3 \times 26 \times 55 = 4,290 \rightarrow P(R3) \approx 0.9\%$

Initially, R2 (which consumes X_1 and X_3 while producing X_2) fires most frequently, causing X_2 to grow rapidly. As X_2 grows, R1 becomes increasingly likely (since its rate is proportional to X_2). R1 produces 4 molecules of X_3 per firing while consuming $2X_1 + X_2$, creating a strong positive feedback loop that drives X_3 upward. Meanwhile:

- For C1 ($X_1 \geq 150$): X_1 starts at 110 and is consumed by both R1 and R2, making X_1 growth extremely unlikely.
- For C2 ($X_2 < 10$): X_2 starts at 26 and R2 produces $3X_2$ per firing, so X_2 tends to grow, not shrink.

The dynamics strongly favor X_3 growth, making $\Pr(C3) \approx 1$, while $\Pr(C1)$ and $\Pr(C2)$ are negligibly small.

Problem 1(b): Mean and Variance After 7 Steps

Setup: Starting from $S = [X_1=9, X_2=8, X_3=7]$, compute the mean and variance of the probability distributions of X_1 , X_2 , and X_3 after exactly 7 reaction steps.

Method: 100,000 independent trials were performed. Each trial ran for exactly 7 reaction steps (no threshold termination). The final values of X_1 , X_2 , and X_3 were recorded, and sample mean and variance were computed.

Results:

Species	Initial	Mean (after 7 steps)	Variance
X_1	9	5.8313	5.9089
X_2	8	12.5082	8.9293
X_3	7	7.7949	8.9469

Analysis:

After 7 steps from $[9, 8, 7]$:

- **X_1 decreases** on average (from 9 to about 5.83). This is because R1 (consuming $2X_1$) and R2 (consuming X_1) are the dominant reactions, while R3 (producing $2X_1$) fires less frequently.
- **X_2 increases** on average (from 8 to about 12.51). R2 produces $3X_2$ per firing and is the most probable reaction, driving X_2 growth.
- **X_3 stays roughly similar** (from 7 to about 7.79). R1 produces $4X_3$ but R2 consumes $2X_3$, and R3 consumes X_3 . The net effect is a slight increase.

Conservation check: The total initial count is $9 + 8 + 7 = 24$. The total final average is $5.83 + 12.51 + 7.79 = 26.13$. The increase of about 2.13 is consistent with R1 (the only reaction that changes total count, adding +1 per firing) firing about 2.13 times on average out of 7 total steps (about 30.4% of firings being R1).

The variances (5.9, 8.9, 8.9) are significant relative to the means, reflecting the inherent stochasticity of the system at these small molecule counts.

Problem 2: Synthesizing Randomness - Lambda Bacteriophage

2.1 Background Theory

The lambda bacteriophage is a virus that infects *E. coli* bacteria and makes a probabilistic choice between two survival strategies:

- **Stealth (lysogeny):** The phage integrates its DNA into the host genome and replicates passively when the bacterium divides. This is controlled by the cI regulatory protein (measured as cI_2 , the dimerized form).
- **Hijack (lysis):** The phage commandeers the host's machinery to produce many copies of itself, ultimately killing the bacterium. This is controlled by the Cro regulatory protein (measured as Cro_2 , the dimerized form).

The decision is governed by a complex genetic regulatory network involving about 60 molecular species and about 118 reactions. The **multiplicity of infection (MOI)** - the number of phage particles infecting a single bacterium - is a key environmental parameter that influences the stealth-vs-hijack decision.

The simulation terminates when either:

- $cI_2 > 145 \rightarrow$ stealth mode
- $Cro_2 > 55 \rightarrow$ hijack mode

2.2 Results

200 trials were performed for each MOI value from 1 to 10:

MOI	P(Stealth: $cI_2 > 145$)	P(Hijack: $Cro_2 > 55$)
1	15.0%	85.0%
2	19.5%	80.5%
3	29.5%	70.5%
4	22.5%	77.5%
5	34.0%	66.0%
6	27.0%	73.0%
7	32.0%	68.0%
8	32.0%	68.0%

MOI	P(Stealth: $cI_2 > 145$)	P(Hijack: $Cro_2 > 55$)
9	37.5%	62.5%
10	36.0%	64.0%

2.3 Analysis

The results reveal a clear trend:

- At **low MOI** (1–2), the phage strongly favors the **hijack strategy** (about 80–85%). With fewer phage genomes competing for the host's resources, lysis is the dominant pathway.
- As **MOI increases** (toward 10), the probability of **stealth mode increases** (up to about 36–38%), while hijack probability decreases.

This makes biological sense: at higher MOI, more phage particles contribute cI protein production. The cI protein represses lytic (hijack) genes and promotes lysogeny (stealth). With more phage genomes cooperating to produce cI, the system is more likely to cross the $cI_2 > 145$ threshold before Cro_2 accumulates past 55.

However, even at MOI = 10, hijack mode still dominates (about 64%). This reflects the inherent bias of the lambda regulatory circuit toward the lytic pathway - stealth requires sustained accumulation of cI_2 to high levels, which is harder to achieve than the Cro_2 threshold.

The probabilities exhibit some noise due to the stochastic nature of the simulation (200 trials per point). More trials would smooth the trend, but the overall pattern is consistent: **increasing MOI shifts the balance toward stealth mode.**

Problem 3: Synthesizing Chemical Reaction Networks

3.1 Background Theory

The synthesis problem asks: given a desired mathematical function, can we design a set of chemical reactions that computes it? The key building blocks provided in the course are:

Multiplication module ($Z = X \times Y$): Consumes X one at a time; for each X consumed, copies all of Y into Z. Uses rate separation to ensure sequential processing:

- Slow: $X \rightarrow a$ (one X consumed per iteration)
- Fast: $a + Y \rightarrow a + Y' + Z$ (a catalytically copies Y)
- Fast: $a \rightarrow \emptyset$ (a decays after copying)

- Medium: $Y' \rightarrow Y$ (restore Y for next iteration)

Logarithm module ($Y = \log_2(X)$): Halves X iteratively; each halving increments the output:

- Slow: $b \rightarrow a + b$ (catalyst drives iterations)
- Fast: $a + 2X \rightarrow c + X' + a$ (halving: consume pairs)
- Fast: $2c \rightarrow c$ (reduce c count to 1)
- Fast: $a \rightarrow \emptyset$ (a decays)
- Medium: $X' \rightarrow X$ (restore halved X)
- Medium: $c \rightarrow Y$ (output: one per halving)

Exponentiation module ($Y = 2^X$): Doubles Y for each X consumed:

- Slow: $X \rightarrow a$ (consume one input per iteration)
- Fast: $a + Y \rightarrow a + 2Y'$ (double Y)
- Fast: $a \rightarrow \emptyset$ (a decays)
- Medium: $Y' \rightarrow Y$ (restore doubled Y)

The correctness of these modules relies on **rate separation**: reactions at different speed tiers complete in a well-defined order within each iteration, preventing interference.

Problem 3(a): $Z_\infty = X_0 \times \log_2(Y_0)$

Design Approach: Rather than chaining separate log and multiply modules (which would require complex synchronization), I designed an **integrated log-multiply module** that computes the product directly. The key insight: in the logarithm module, each halving iteration produces 1 unit of output. If instead of outputting 1, we output X copies per halving, the result is $X \times \log_2(Y)$.

Chemical Reaction Network:

Reaction	Rate	Role
$b \rightarrow a_1 + b$	1	Slow catalyst: drives halving iterations
$a_1 + 2Y \rightarrow c + Y' + a_1$	10^7	Fastest: halve Y (catalytic in a_1)
$2c \rightarrow c$	10^7	Fastest: reduce c to exactly 1
$a_1 \rightarrow W_1$	10^5	Fast: a_1 decays after halving

Reaction	Rate	Role
$c + X \rightarrow c + X' + Z$	10^6	Medium-fast: c catalytically copies X into Z
$c \rightarrow W_2$	10^4	Medium: c decays after copying
$X' \rightarrow X$	100	Slow: restore X for next iteration
$Y' \rightarrow Y$	100	Slow: restore halved Y for next iteration

How it works (trace with $X=5$, $Y=16$):

Each iteration of the $b \rightarrow a_1 + b$ catalyst:

1. a_1 halves Y by consuming pairs: $Y=16 \rightarrow 8$ pairs consumed $\rightarrow c=8$, $Y'=8$, $Y=0$
2. $2c \rightarrow c$ reduces: $c=8 \rightarrow 4 \rightarrow 2 \rightarrow 1$
3. a_1 decays to W_1
4. The single c molecule copies all of X into Z: $Z += 5$
5. c decays to W_2
6. $X' \rightarrow X$ restores X to 5; $Y' \rightarrow Y$ restores Y to 8

Repeating: Y goes $16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$ (four halvings), each time adding 5 to Z.

Final: $Z = 5 \times 4 = 20 = X \times \log_2(Y)$. ✓

Simulation Results ($X=5$, $Y=16$, 1000 trials, time limit=100):

Species	Initial	Mean	Variance
Y	16	1.00	0.00
X	5	5.00	0.00
Z	0	20.32	1.74

The mean of $Z = 20.32$, very close to the expected value of $20 = 5 \times \log_2(16)$.

The small upward bias (about 0.32) and variance (about 1.74) arise from stochastic effects: occasionally, race conditions in the rate hierarchy cause an extra partial copy or an extra halving iteration. The variance is small relative to the mean, confirming the CRN computes the desired function reliably.

Problem 3(b): $Y_\infty = 2^{(\log_2(X_0))}$

Design Approach: This computation chains two modules sequentially:

1. **Log module:** Computes $L = \log_2(X)$ by iteratively halving X
2. **Exp module:** Computes $Y = 2^L$ by doubling Y for each L consumed

The two modules are connected through the intermediate species L . Rate separation ensures the log module (slow trigger rate = 1) completes well before the exp module (very slow trigger rate = 0.01) consumes significant L .

Chemical Reaction Network:

Log module ($L = \log_2(X)$):

Reaction	Rate	Role
$b \rightarrow a_1 + b$	1	Slow catalyst: drives halving iterations
$a_1 + 2X \rightarrow c + X' + a_1$	10^7	Fastest: halve X
$2c \rightarrow c$	10^7	Fastest: reduce c to 1
$a_1 \rightarrow W_1$	10^5	Fast: a_1 decays
$X' \rightarrow X$	100	Medium: restore halved X
$c \rightarrow L$	100	Medium: output one L per halving

Exp module ($Y = 2^L$):

Reaction	Rate	Role
$L \rightarrow a_2$	0.01	Very slow: consume one L per doubling
$a_2 + Y \rightarrow a_2 + 2Y'$	10^7	Fastest: a_2 catalytically doubles Y
$a_2 \rightarrow W_2$	10^5	Fast: a_2 decays after doubling
$Y' \rightarrow Y$	1000	Medium: restore doubled Y

How it works (trace with $X=8$, $Y=1$):

Log phase (b drives iterations, each halving X):

- $X = 8 \rightarrow 4 \rightarrow 2 \rightarrow 1$ (three halvings), producing $L = 3$

Exp phase (each L consumed doubles Y):

- $L=3$: consume one L , $Y=1 \rightarrow 2$

- $L=2$: consume one L, $Y=2 \rightarrow 4$
- $L=1$: consume one L, $Y=4 \rightarrow 8$

Final: $Y = 2^3 = 8 = X_0$. ✓

Simulation Results ($X=8$, Y initially 1, 1000 trials, time limit=500):

Species	Initial	Mean	Variance
X	8	1.00	0.00
L	0	0.02	0.02
Y	1	8.21	3.37

The mean of $Y = 8.21$, very close to the expected value of $8 = 2^{(\log_2(8))} = X_0$.

The residual $L \approx 0.02$ indicates that in about 2% of trials, one L molecule was not consumed before the time limit. The slight upward bias in Y (8.21 vs 8) comes from rare stochastic events where an extra halving produces an extra L, leading to an extra doubling. The variance (3.37) reflects the inherent stochasticity of the rate-separation approach.